

1)a) To find the eigenvalues and corresponding eigenvectors for the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix},$$

we will follow these steps: Step 1: Eigenvalues The eigenvalues of a diagonal matrix are the diagonal entries. Therefore, the eigenvalues for this matrix are:

$$\lambda_1 = 2, \quad \lambda_2 = 3, \quad \lambda_3 = 5.$$

Step 2: Eigenvectors For each eigenvalue, the corresponding eigenvector can be found by solving $(A - \lambda I)\mathbf{v} = 0$, where I is the identity matrix and λ is the eigenvalue.

For $\lambda_1 = 2$: We need to solve $(A - 2I)\mathbf{v} = 0$:

$$A - 2I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Solving $(A - 2I)\mathbf{v} = 0$:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

From this, we get:

$$v_2 = 0, \quad v_3 = 0.$$

Thus, the eigenvector corresponding to $\lambda_1 = 2$ is:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

For $\lambda_2 = 3$: We need to solve $(A - 3I)\mathbf{v} = 0$:

$$A - 3I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Solving $(A - 3I)\mathbf{v} = 0$:

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

From this, we get:

$$v_1 = 0, \quad v_3 = 0.$$

Thus, the eigenvector corresponding to $\lambda_2 = 3$ is:

$$\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

For $\lambda_3 = 5$: We need to solve $(A - 5I)\mathbf{v} = 0$:

$$A - 5I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solving $(A - 5I)\mathbf{v} = 0$:

$$\begin{bmatrix} -3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

From this, we get:

$$v_1 = 0, \quad v_2 = 0.$$

Thus, the eigenvector corresponding to $\lambda_3 = 5$ is:

$$\mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Final Answer: - Eigenvalue $\lambda_1 = 2$ has the eigenvector $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. - Eigenvalue $\lambda_2 = 3$ has the eigenvector

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \text{ - Eigenvalue } \lambda_3 = 5 \text{ has the eigenvector } \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

1)b) To find the eigenvalues and corresponding eigenvectors for the matrix

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{bmatrix},$$

we will follow these steps:

Step 1: Eigenvalues We need to find the determinant of $A - \lambda I$ and solve for λ :

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & -1 & 2 \\ 2 & -2 - \lambda & 4 \\ 3 & -3 & 6 - \lambda \end{bmatrix}.$$

The determinant of this matrix is:

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & -1 & 2 \\ 2 & -2 - \lambda & 4 \\ 3 & -3 & 6 - \lambda \end{vmatrix}.$$

We can compute the determinant by expanding along the first row:

$$\det(A - \lambda I) = (1 - \lambda) \begin{vmatrix} -2 - \lambda & 4 \\ -3 & 6 - \lambda \end{vmatrix} - (-1) \begin{vmatrix} 2 & 4 \\ 3 & 6 - \lambda \end{vmatrix} + 2 \begin{vmatrix} 2 & -2 - \lambda \\ 3 & -3 \end{vmatrix}.$$

After calculating, the characteristic polynomial simplifies to:

$$\det(A - \lambda I) = -\lambda^3 + 5\lambda^2 = -\lambda^2(\lambda - 5).$$

Thus, the eigenvalues are:

$$\lambda_1 = 0, \quad \lambda_2 = 5.$$

Step 2: Eigenvectors For each eigenvalue, the corresponding eigenvector can be found by solving $(A - \lambda I)\mathbf{v} = 0$, where I is the identity matrix and λ is the eigenvalue.

For $\lambda_1 = 0$: We need to solve $A\mathbf{v} = 0$:

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{bmatrix}.$$

Solving the system of equations:

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

we find that the eigenvector corresponding to $\lambda_1 = 0$ is:

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

For $\lambda_2 = 5$: We need to solve $(A - 5I)\mathbf{v} = 0$:

$$A - 5I = \begin{bmatrix} -4 & -1 & 2 \\ 2 & -7 & 4 \\ 3 & -3 & 1 \end{bmatrix}.$$

Solving the system of equations:

$$\begin{bmatrix} -4 & -1 & 2 \\ 2 & -7 & 4 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

we find that the eigenvector corresponding to $\lambda_2 = 5$ is:

$$\mathbf{v}_2 = \begin{bmatrix} \frac{-1}{2} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

Final Answer: - Eigenvalue $\lambda_1 = 0$ has the eigenvector $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. - Eigenvalue $\lambda_2 = 5$ has the eigenvector

$$\begin{bmatrix} \frac{-1}{2} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

To find the eigenvalues and corresponding eigenvectors for the following matrices:

2)a). Matrix:

$$A = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$$

Eigenvalues: To find the eigenvalues, solve the characteristic equation $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \det \begin{bmatrix} 1 - \lambda & -i \\ i & 1 - \lambda \end{bmatrix}$$

The determinant is:

$$(1 - \lambda)(1 - \lambda) - (-i)(i) = (1 - \lambda)^2 + 1 = 0$$

$$(1 - \lambda)^2 = -1 \Rightarrow 1 - \lambda = \pm i \Rightarrow \lambda = 1 \pm i$$

Thus, the eigenvalues are $\lambda_1 = 1 + i$ and $\lambda_2 = 1 - i$.

Eigenvectors: For $\lambda_1 = 1 + i$, solve $(A - (1 + i)I)\mathbf{v} = 0$:

$$A - (1 + i)I = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} - \begin{bmatrix} 1 + i & 0 \\ 0 & 1 + i \end{bmatrix} = \begin{bmatrix} -i & -i \\ i & -i \end{bmatrix}$$

Solving $\begin{bmatrix} -i & -i \\ i & -i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, we get:

$$v_1 = v_2 \quad \Rightarrow \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 1 - i$, solve $(A - (1 - i)I)\mathbf{v} = 0$:

$$A - (1 - i)I = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} - \begin{bmatrix} 1-i & 0 \\ 0 & 1-i \end{bmatrix} = \begin{bmatrix} i & -i \\ i & i \end{bmatrix}$$

Solving $\begin{bmatrix} i & -i \\ i & i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, we get:

$$v_1 = -v_2 \quad \Rightarrow \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

b). Matrix:

$$B = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Eigenvalues: Solve $\det(B - \lambda I) = 0$:

$$\det \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \det \begin{bmatrix} -\lambda & -i \\ i & -\lambda \end{bmatrix}$$

The determinant is:

$$(-\lambda)(-\lambda) - (-i)(i) = \lambda^2 + 1 = 0$$

$$\lambda^2 = -1 \quad \Rightarrow \quad \lambda = \pm i$$

Thus, the eigenvalues are $\lambda_1 = i$ and $\lambda_2 = -i$.

Eigenvectors: For $\lambda_1 = i$, solve $(B - iI)\mathbf{v} = 0$:

$$B - iI = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} - \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} -i & -i \\ i & -i \end{bmatrix}$$

Solving $\begin{bmatrix} -i & -i \\ i & -i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, we get:

$$v_1 = -v_2 \quad \Rightarrow \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

For $\lambda_2 = -i$, solve $(B + iI)\mathbf{v} = 0$:

$$B + iI = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} + \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} i & -i \\ i & i \end{bmatrix}$$

Solving $\begin{bmatrix} i & -i \\ i & i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, we get:

$$v_1 = v_2 \quad \Rightarrow \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

c). Matrix:

$$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Eigenvalues: Solve $\det(C - \lambda I) = 0$:

$$\det \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \det \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & 0 & -\lambda \end{bmatrix}$$

The determinant is:

$$-\lambda \begin{vmatrix} -\lambda & 1 \\ 0 & -\lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 1 & -\lambda \end{vmatrix} = -\lambda(\lambda^2) - 1(-\lambda - 1)$$

$$-\lambda^3 + \lambda + 1 = 0$$

Thus, the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\lambda_3 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$.
Eigenvectors: For $\lambda_1 = 1$, solve $(C - I)\mathbf{v} = 0$:

$$C - I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

Solving $\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, we get:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

We are tasked with finding the algebraic and geometric multiplicities of the eigenvalues of the following matrices, and determining whether they are diagonalizable.

3)a). Matrix A :

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Eigenvalues: To find the eigenvalues, we solve the characteristic equation $\det(A - \lambda I) = 0$:

$$A - \lambda I = \begin{bmatrix} 0 - \lambda & 1 & 0 \\ 1 & 0 - \lambda & 1 \\ 0 & 1 & 0 - \lambda \end{bmatrix} = \begin{bmatrix} -\lambda & 1 & 0 \\ 1 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{bmatrix}$$

The determinant is:

$$\det(A - \lambda I) = -\lambda \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 0 & -\lambda \end{vmatrix}$$

$$= -\lambda(\lambda^2 - 1) - 1(-\lambda) = -\lambda^3 + \lambda + \lambda = -\lambda^3 + 2\lambda$$

$$= \lambda(\lambda^2 - 2)$$

So the eigenvalues are $\lambda_1 = 0$, $\lambda_2 = \sqrt{2}$, and $\lambda_3 = -\sqrt{2}$.

Algebraic Multiplicities: The algebraic multiplicities of the eigenvalues are: - $\lambda_1 = 0$ has algebraic multiplicity 1, - $\lambda_2 = \sqrt{2}$ has algebraic multiplicity 1, - $\lambda_3 = -\sqrt{2}$ has algebraic multiplicity 1.

Geometric Multiplicities: To find the geometric multiplicities, we solve $(A - \lambda I)\mathbf{v} = 0$ for each eigenvalue.

- For $\lambda_1 = 0$, solve $A\mathbf{v} = 0$. The null space of A gives the geometric multiplicity, which is 1 because the null space consists of only one linearly independent vector.

- For $\lambda_2 = \sqrt{2}$ and $\lambda_3 = -\sqrt{2}$, solving $(A - \lambda I)\mathbf{v} = 0$ for each eigenvalue results in one linearly independent eigenvector for each, so their geometric multiplicities are both 1.

Diagonalizability: Since the algebraic multiplicities are equal to the geometric multiplicities for all eigenvalues, the matrix A is diagonalizable.

3)b) Matrix B :

$$B = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{bmatrix}$$

Eigenvalues: To find the eigenvalues, we solve the characteristic equation $\det(B - \lambda I) = 0$:

$$B - \lambda I = \begin{bmatrix} 1 - \lambda & -1 & 0 \\ 1 & 2 - \lambda & -1 \\ 3 & 2 & -2 - \lambda \end{bmatrix}$$

The determinant is:

$$\det(B - \lambda I) = (1 - \lambda) \begin{vmatrix} 2 - \lambda & -1 \\ 2 & -2 - \lambda \end{vmatrix} - (-1) \begin{vmatrix} 1 & -1 \\ 3 & -2 - \lambda \end{vmatrix}$$

Expanding the determinant:

$$\begin{aligned} &= (1 - \lambda) ((2 - \lambda)(-2 - \lambda) - (-1)(2)) + \begin{vmatrix} 1 & -1 \\ 3 & -2 - \lambda \end{vmatrix} \\ &= (1 - \lambda) (\lambda^2 - 4) + (2 + \lambda) = (1 - \lambda)(\lambda^2 - 4) + 2 + \lambda \end{aligned}$$

After solving the cubic equation, we find that the eigenvalues are $\lambda_1 = 3$, $\lambda_2 = -1$, and $\lambda_3 = -1$.

Algebraic Multiplicities: The algebraic multiplicities of the eigenvalues are: - $\lambda_1 = 3$ has algebraic multiplicity 1, - $\lambda_2 = -1$ has algebraic multiplicity 2.

Geometric Multiplicities: - For $\lambda_1 = 3$, solving $(B - 3I)\mathbf{v} = 0$ gives a geometric multiplicity of 1. - For $\lambda_2 = -1$, solving $(B + I)\mathbf{v} = 0$ gives a geometric multiplicity of 1.

Diagonalizability: Since the geometric multiplicity of $\lambda_2 = -1$ is less than its algebraic multiplicity, matrix B is **not** diagonalizable.

4. We are tasked with finding the characteristic polynomial, minimal polynomial, and determining whether the matrices are diagonalizable. Below are the solutions for each matrix.

1. Matrix A :

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Characteristic Polynomial: The characteristic polynomial is given by the determinant of $A - \lambda I$:

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 1 & 0 \\ 0 & 4 - \lambda & 0 \\ 0 & 0 & 2 - \lambda \end{bmatrix}$$

The determinant is:

$$\begin{aligned} \det(A - \lambda I) &= (4 - \lambda) \cdot \begin{vmatrix} 4 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} \\ &= (4 - \lambda) \cdot (4 - \lambda) \cdot (2 - \lambda) \end{aligned}$$

Thus, the characteristic polynomial is:

$$p_A(\lambda) = (4 - \lambda)^2 \cdot (2 - \lambda)$$

Minimal Polynomial: The minimal polynomial is determined by examining the matrix A . Since A is already in upper triangular form, the eigenvalues of A are $\lambda_1 = 4$ (with algebraic multiplicity 2) and $\lambda_2 = 2$ (with algebraic multiplicity 1). The minimal polynomial is formed by the distinct eigenvalues:

$$m_A(\lambda) = (\lambda - 4) \cdot (\lambda - 2)$$

Diagonalizability: Since the algebraic multiplicity of $\lambda_1 = 4$ is 2, and the geometric multiplicity (number of independent eigenvectors) is also 2, matrix A is ****diagonalizable****.

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4.b) Matrix B :

$$B = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

Characteristic Polynomial: The characteristic polynomial is given by the determinant of $B - \lambda I$:

$$B - \lambda I = \begin{bmatrix} 3 - \lambda & 1 & 0 \\ 0 & 3 - \lambda & 2 \\ 0 & 0 & 3 - \lambda \end{bmatrix}$$

The determinant is:

$$\begin{aligned} \det(B - \lambda I) &= (3 - \lambda) \cdot \begin{vmatrix} 3 - \lambda & 2 \\ 0 & 3 - \lambda \end{vmatrix} \\ &= (3 - \lambda) \cdot (3 - \lambda) \cdot (3 - \lambda) \end{aligned}$$

Thus, the characteristic polynomial is:

$$p_B(\lambda) = (3 - \lambda)^3$$

Minimal Polynomial: Since the eigenvalue $\lambda_1 = 3$ has algebraic multiplicity 3, and we have a single eigenvalue for B , the minimal polynomial must be of the form:

$$m_B(\lambda) = (\lambda - 3)^2$$

Diagonalizability: Matrix B has a single eigenvalue with algebraic multiplicity 3, but the geometric multiplicity is 1 (since there is only one independent eigenvector corresponding to $\lambda = 3$). Therefore, matrix B is ****not diagonalizable****.

⑤ a) $A = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$

char. poly, $A - \lambda I = 0$
 $\Rightarrow \begin{bmatrix} 5-\lambda & 1 \\ 0 & 5-\lambda \end{bmatrix} = 0$
 $\Rightarrow \begin{bmatrix} 5-\lambda & 1 \\ 0 & 5-\lambda \end{bmatrix} = 0$
 $\Rightarrow 25 - 10\lambda + \lambda^2 = 0$

Using Cayley-Hamilton theorem,

$$25 - 10A + A^2 = 0$$

$$\Rightarrow A(A - 10) = -25$$

$$\Rightarrow A^{-1} = -\frac{1}{25} (A - 10)$$

$$= \frac{1}{25} (10I - A)$$

⑥ $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 5 & -4 \\ 0 & 2 & 5 \end{bmatrix}$

$$A - \lambda I = 0 \Rightarrow (1-\lambda)(\lambda^2 - 10\lambda + 33) = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 33 = \lambda(\lambda^2 - 10\lambda + 33)$$

Using Cayley-Hamilton theorem,

$$-A^3 + 11A^2 - 43A + 33 = 0$$

$$\Rightarrow A(A^2 - 11A + 43I) = 33$$

$$\Rightarrow A^{-1} = \frac{1}{33} (A^2 - 11A + 43I)$$

⑦ $A = \begin{bmatrix} 3 & 1 & 0 \\ -3 & 5 & 3 \\ 0 & 3 & 3 \end{bmatrix}$

$$A - \lambda I = 0 \Rightarrow -\lambda^3 + 11\lambda^2 - 33\lambda + 27 = 0$$

Using Cayley-Hamilton theorem,

$$-A^3 + 11A^2 - 33A + 27 = 0$$

$$\Rightarrow A(A^2 - 11A + 33I) = 27$$

$$\Rightarrow A^{-1} = \frac{1}{27} (A^2 - 11A + 33I)$$